Game Theory for Strategic Business Decisions

Think Like Your Competitors Think

Economics for Business Decisions
PGDM Batch 2025–27

What is Game Theory?

"The science of strategic thinking"

- Mathematical framework for analyzing strategic interactions
- Predicts outcomes when multiple decision-makers interact
- Your success depends on others' choices (and vice versa)

Key Question: What should you do when what's best for you depends on what others do?

Why MBA Students Must Master Game Theory

Career Applications:

- **Pricing Strategy:** How to respond to competitor price cuts
- **Market Entry:** When to enter/exit markets
- **Negotiations:** Salary, contracts, partnerships
- **Product Launch:** Timing and positioning decisions
- **Investment:** Portfolio strategies and risk management

Reality Check: Every major business decision involves game theory!

The Game Theory Toolkit

1. **Players:** Decision makers (companies, individuals)
2. **Strategies:** Available choices/actions
3. **Payoffs:** Outcomes/profits from each combination
4. **Information:** What each player knows
5. **Rules:** When and how decisions are made

Goal: Find the **Nash Equilibrium** - where no player wants to change strategy given others' choices

The Famous Prisoner's Dilemma

Two suspects arrested, held separately, can't communicate

		Prisoner B	
		Confess	Stay Silent
2***Prisoner A**	Confess	(-5, -5)	(0, -10)
	Stay Silent	(-10, 0)	(-1, -1)

Nash Equilibrium: Both confess (-5, -5)
Optimal Outcome: Both stay silent (-1, -1)
The Tragedy: Individual rationality collective optimality

Business Prisoner's Dilemma: Price Wars

Case Study: Jio vs Airtel (2016-2018)

		Airtel	
		High Price	Low Price
2***Jio**	High Price	(500, 400)	(100, 600)
	Low Price	(300, 50)	(200, 200)

Reality: Both chose low prices $\rightarrow$ Industry profits collapsed
Winner: Consumers got cheaper data/calls
Lesson: Sometimes you must play the "bad" equilibrium

Breaking Out of Prisoner's Dilemmas

Real Strategies Companies Use:

- **Repeated Games:** "If you cut prices, I'll cut mine forever"
- **Communication:** Industry associations, informal meetings
- **Commitment:** Public announcements that are hard to reverse
- **Product Differentiation:** Avoid direct competition
- **Market Segmentation:** Focus on different customer segments

Example: Airlines avoid price wars by differentiating (Premium, Budget, Regional)

Coordination Game: The Battle of Standards

When players benefit from doing the same thing
Case: UPI vs Wallets (2016-2020)

- **Players:** Payment companies (Paytm, PhonePe, GPay)
- **Choice:** Proprietary wallets vs UPI standard
- **Payoffs:** Network effects - more users = more value

Multiple Equilibria:

- Everyone uses wallets $\rightarrow$ Fragmented market
- Everyone uses UPI $\rightarrow$ Unified ecosystem

Result: UPI won $\rightarrow$ 100+ lakh crore transaction volume

Sequential Games: First Mover Advantage

Decisions made in sequence, not simultaneously

Case Study: Amazon vs Flipkart in India

- **2007:** Flipkart enters first $\rightarrow$ Established brand
- **2013:** Amazon enters $\rightarrow$ Must decide: Compete or Exit?
- **Amazon's Strategy:** Massive investment (20,000+ crores)
- **Flipkart's Response:** Match investment or lose market share

Game Tree Analysis: Early entry creates strategic advantages but also commitment costs

Entry Deterrence: The Credible Threat

How incumbents keep competitors out
Strategy Options:

1. **Price Predation:** Cut prices below cost
2. **Capacity Building:** Over-invest to signal commitment
3. **Product Proliferation:** Fill all market niches
4. **Switching Costs:** Make customers sticky

Real Example: Google's free services create switching costs
Key Insight: Threat must be credible and profitable to execute

Mixed Strategies: The Art of Unpredictability

When you should randomize your decisions
Business Applications:

- **Auditing:** Random checks are more effective than predictable ones
- **Product Launches:** Surprise timing prevents competitive response
- **Pricing:** Occasional sales keep competitors guessing
- **Hiring:** Unpredictable interview processes

Example: Ola's surge pricing algorithm uses mixed strategies to optimize driver supply

Zero-Sum vs Non-Zero-Sum Games

Zero-Sum:

- One player's gain = other's loss
- Total payoffs = 0
- Pure competition
- Example: Market share battles

Non-Zero-Sum:

- Win-win possibilities exist
- Total value can be created
- Cooperation beneficial
- Example: Strategic partnerships

MBA Insight: Always look for ways to make games non-zero-sum through innovation, partnerships, or market expansion

Information Asymmetry: The Knowledge Game

When players have different information
Types:

- **Hidden Information:** One player knows something others don't
- **Hidden Actions:** Actions are unobservable
- **Signaling:** Actions that reveal information
- **Screening:** Actions that elicit information

Real Example: Job interviews are signaling games - candidates signal quality through education, experience, and communication

Auction Theory: Strategic Bidding

Different Auction Types:

1. **English Auction:** Open ascending bids (eBay)
2. **Dutch Auction:** Descending price (flower markets)
3. **First-Price Sealed Bid:** Highest bid wins, pays bid
4. **Second-Price Sealed Bid:** Highest bid wins, pays second-highest

Business Applications:

- IPO pricing
- Government contracts
- Spectrum auctions
- Real estate

Case Study 1: The Great Indian Telecom War

Players: Jio, Airtel, Vodafone-Idea
Game: Pricing and investment strategies (2016-2025)
Round 1 (2016): Jio's free services
Round 2 (2017): Competitors match prices
Round 3 (2018-20): Consolidation (Vodafone-Idea merger)
Round 4 (2021-25): 5G investment race
Game Theory Lessons:

- Aggressive entry can reshape entire industries
- Financial strength affects strategic options
- Government policy changes game rules

Case Study 2: COVID-19 Vaccine Distribution

Players: Countries, pharmaceutical companies
Game: Resource allocation and cooperation
Strategies:

- **Vaccine Nationalism:** Hoard doses for own population
- **Global Cooperation:** Share resources equitably
- **Diplomatic Tool:** Use vaccines for soft power

India's Strategy: Initially export (Vaccine Maitri), then prioritize domestic needs during Delta wave
Lesson: Global challenges require balancing self-interest with collective action

Case Study 3: Streaming Wars in India

Players: Netflix, Amazon Prime, Disney+ Hotstar, Jio Cinema
Game: Content strategy and pricing

Platform	Strategy	Price	Content Focus
Netflix	Premium Global	199-649	Original Series
Prime	Bundle with Shopping	179/year	Mixed Content
Hotstar	Sports + Local	299/year	Cricket + Regional
JioCinema	Free with Data	Free	Bollywood + Sports

Nash Equilibrium: Each platform finds its niche rather than direct competition

Interactive Exercise: The Startup Dilemma

Scenario: You're launching a food delivery startup. Zomato and Swiggy dominate the market.

Your Options:

1. Compete directly with aggressive pricing
2. Focus on a niche (healthy food, tier-2 cities)
3. Partner with one of the incumbents
4. Exit and pursue different opportunity

Consider:

- How will Zomato/Swiggy respond?
- What are your resource constraints?
- Can you change the game rules?

Discuss in pairs for 3 minutes

Advanced Game Theory Concepts

Evolutionary Game Theory:

- Strategies evolve over time
- Learning and adaptation
- Example: Business model evolution in e-commerce

Behavioral Game Theory:

- Real people don't always play rationally
- Emotions, biases, and social preferences matter
- Example: Fairness in salary negotiations

Mechanism Design:

- Design rules to achieve desired outcomes
- Example: Incentive systems in organizations

Game Theory in Negotiations

Key Principles:

1. **BATNA:** Best Alternative to Negotiated Agreement
2. **Reservation Price:** Your walk-away point
3. **Zone of Possible Agreement:** Overlap between parties
4. **Anchoring:** First offer influences final outcome
5. **Commitment:** Making credible promises/threats

MBA Application: Salary negotiations, vendor contracts, partnership deals, merger discussions

Common Strategic Mistakes

1. **Assuming opponents are irrational:** They have their own logic
2. **Fighting yesterday's war:** Game rules change constantly
3. **Ignoring second-order effects:** Your move triggers responses
4. **Over-relying on dominant strategies:** Context matters
5. **Forgetting about coalitions:** Multiple players can team up

Remember: Game theory is about strategic thinking, not rigid formulas

Platform Economics & Network Effects

Two-sided markets create unique strategic dynamics
Example: Uber's Strategy

- **Chicken-and-egg problem:** Need drivers to attract riders, need riders to attract drivers
- **Solution:** Subsidize both sides initially
- **Network effects:** More participants = more value for everyone
- **Winner-take-most:** Dominant platforms capture most value

Strategic Implications: First-mover advantage, importance of scale, difficulty of competition

Game Theory in Investment Decisions

Market as a Multiplayer Game:

- **Information asymmetry:** Some investors know more
- **Herding behavior:** Following others' investment decisions
- **Contrarian strategies:** Doing opposite of the crowd
- **Market timing:** Strategic entry and exit

Example: Warren Buffett's strategy - "Be greedy when others are fearful, fearful when others are greedy"
Lesson: Understanding crowd psychology is as important as financial analysis

Technology Adoption Games

Network Externalities in Tech:

- **Standard Wars:** VHS vs Betamax, iOS vs Android
- **Platform Strategies:** Create ecosystems that lock in users
- **Switching Costs:** Make it expensive to leave
- **Compatibility Decisions:** Open vs proprietary standards

Current Example: Electric vehicle charging standards - Tesla vs CCS vs CHAdeMO

Strategic Insight: Sometimes inferior technology wins due to strategic play

Practical Game Theory Checklist

Before making any strategic decision, ask:

1. Who are the key players?
2. What are their possible strategies?
3. How will they likely respond to my actions?
4. What information do they have that I don't?
5. Can I change the game structure to my advantage?
6. Are there opportunities for cooperation?
7. What are the long-term implications?
8. How might new players enter the game?

Key Takeaways for MBA Professionals

1. **Think strategically:** Your success depends on others' decisions
2. **Understand incentives:** What motivates your competitors/partners?
3. **Look for win-win:** Convert zero-sum games to positive-sum when possible
4. **Build credibility:** Your reputation affects your strategic options
5. **Information is power:** Invest in market intelligence
6. **Timing matters:** Know when to move first vs wait
7. **Expect responses:** Every action triggers reactions

Assignment: Game Theory Analysis

Choose any recent business situation and analyze it using game theory:

- Identify players, strategies, and payoffs
- Find Nash equilibrium
- Suggest alternative strategies
- Consider information asymmetries
- Predict future moves

Suggested Topics:

- EV market entry strategies
- Quick commerce competition
- Social media platform wars
- Fintech vs traditional banking

Due: Next class session

Questions & Discussion

Think Strategically, Act Decisively

What strategic situations are you currently facing in your career or business ideas?

How can game theory help you make better decisions?